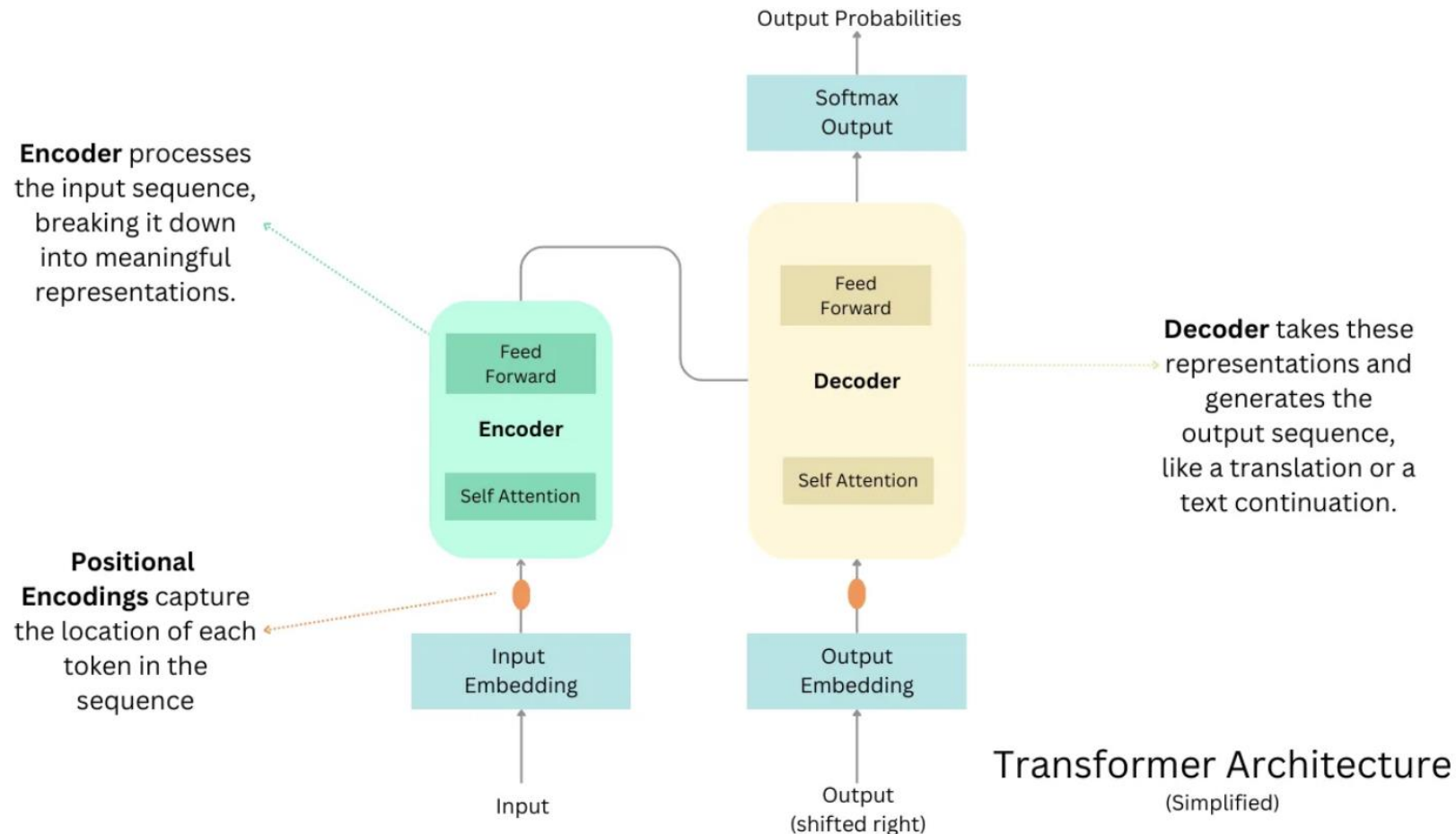


Vision Transformers

Limitations of CNN

- Captures local features only, not global features
- Stacks more number of layers to increase receptive field – potentially leads to vanishing gradient problem
- Translation invariance/equivariance (features are recognized regardless of location)
- Context Understanding (e.g. location of eyes and mouth on the face)

Transformer Architecture



<https://medium.com/@theaveragegal/transformer-architecture-simplified-3fb501d461c8>

Vaswani et al, Attention is All you need, Google Brain Team, 2017

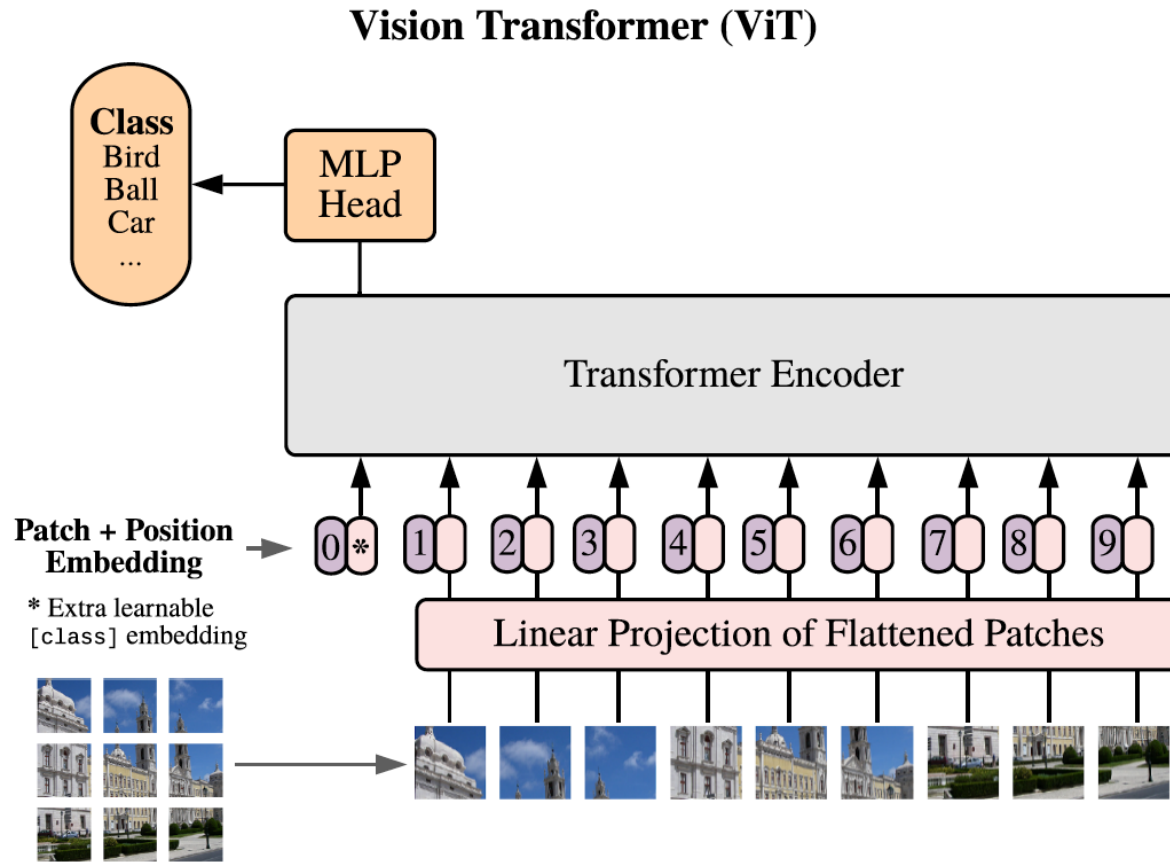
Vision Transformer (ViT)

- Alexey Dosovitskiy et al., An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, Google Brain Team, 2020, <https://arxiv.org/abs/2010.11929v2>
- An image classification model that employs a Transformer-like architecture over patches of the image.
- An image is split into fixed-size patches, each of them are then linearly embedded, position embeddings are added, and the resulting sequence of vectors is fed to a standard Transformer encoder.
- In order to perform classification, the standard approach of adding an extra learnable “classification token” to the sequence is used.

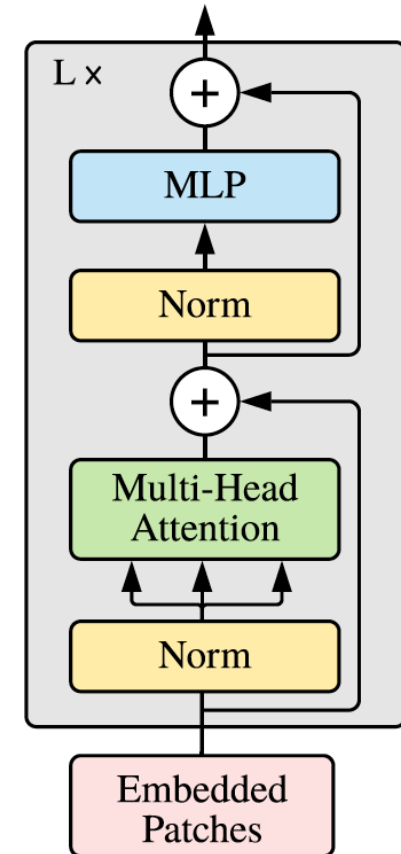
ViT Basics



ViT Architecture



Transformer Encoder



- The cat sat on the mat which was dry

ViT Model Training

- Pretrained on dataset A and finetuned on dataset b.
- Pretraining datasets
 - Imagenet (1k classes, 1.3M images)
 - Imagenet21k (21k classes, 14M images)
 - JFT (18k classes, 303M images)
- Pretrained with Adam optimizer with batch size of 4096 and weight decay of 0.1
- ViT performed better with JFT dataset as compared to Resnet model

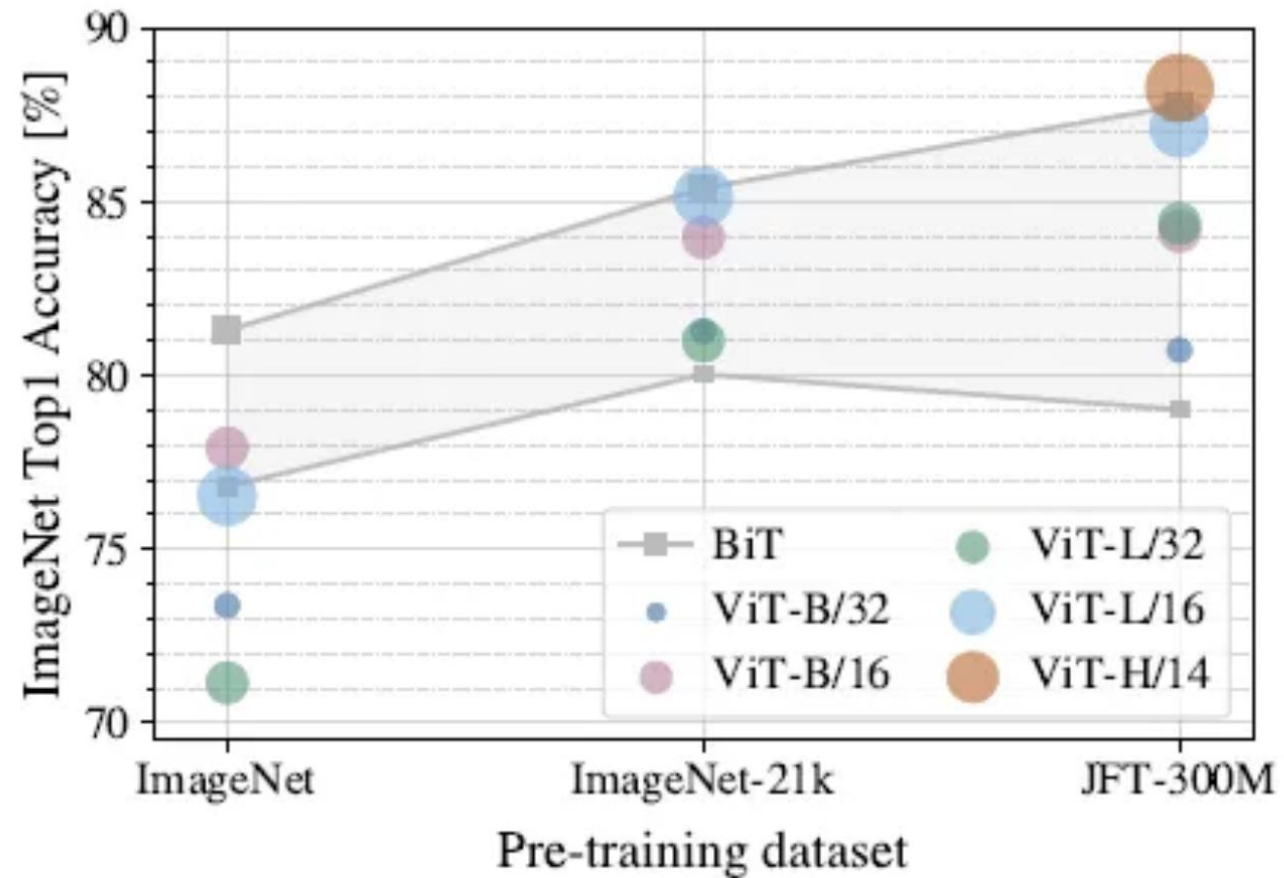
ViT Performance Evaluation

- Different ViT architecture used for performance evaluation

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

- Fine-tuned ViT models on a smaller dataset using SGD with momentum and batch sizes of 512 and 518 for ViT-L/16 and ViT-H/14 models respectively

ViT Performance Evaluation



ViT Performance Evaluation

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21k (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Table 2: Comparison with state of the art on popular image classification benchmarks. We report mean and standard deviation of the accuracies, averaged over three fine-tuning runs. Vision Transformer models pre-trained on the JFT-300M dataset outperform ResNet-based baselines on all datasets, while taking substantially less computational resources to pre-train. ViT pre-trained on the smaller public ImageNet-21k dataset performs well too. *Slightly improved 88.5% result reported in [Touvron et al. \(2020\)](#).

Why ViTs work?

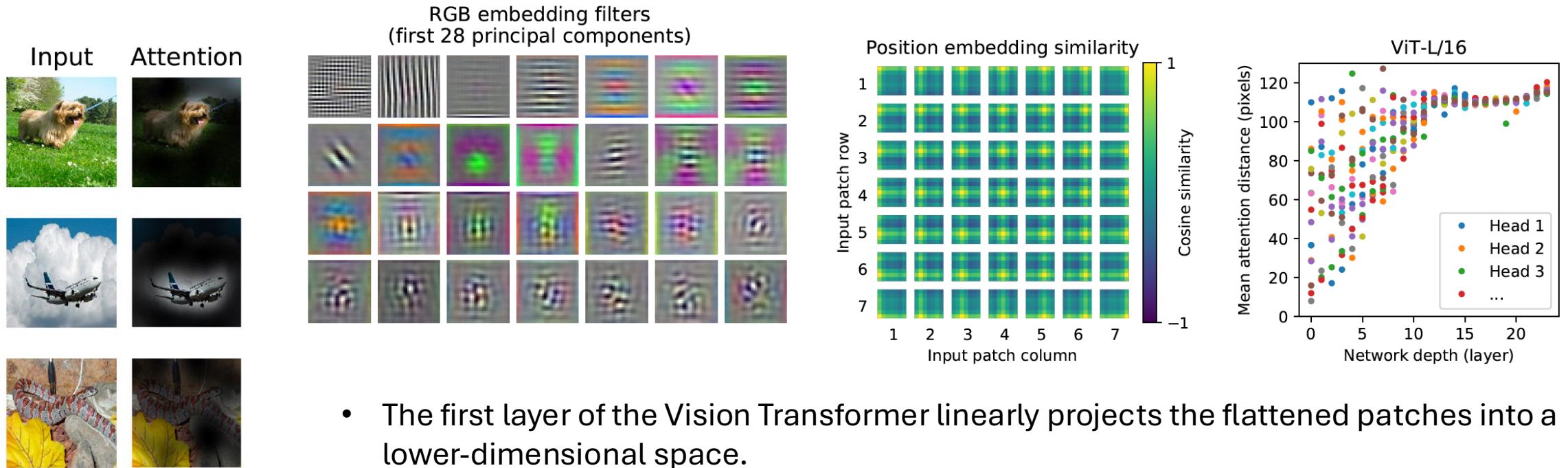


Figure 6: Representative examples of attention from the output token to the input space. See Appendix [D.7](#) for details.

- The first layer of the Vision Transformer linearly projects the flattened patches into a lower-dimensional space.
- After the projection, a learned position embedding is added to the patch representations. The model learns to encode distance within the image in the similarity of position embeddings, i.e. closer patches tend to have more similar position embeddings.
- Self-attention allows ViT to integrate information across the entire image even in the lowest layers. Attention distance is calculated for that. It is analogous to receptive field size in CNNs.

Variations of ViT

- Swin Transformer: Hierarchical Vision Transformer – uses shifted self-attention to focus on small objects in the image
- DINO: Utilises a variant of the traditional knowledge distillation method and applies it to the powerful vision transformer (ViT) architecture.
- TimeSformer: It adapts the standard Transformer architecture to video by enabling spatiotemporal feature learning directly from a sequence of frame-level patches.

References

- Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).
- Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani et al. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv preprint arXiv:2010.11929* (2020).
- <https://www.geeksforgeeks.org/deep-learning/vision-transformer-vit-architecture/>